

Introduction to the Lab

Students will explore pfsense firewall and Snort Intrusion Detection System (IDS) in this lab. Snort IDS, a signature-based intrusion detection system used to detect network threats. Snort can also function as a simple packet logger. The students will utilize snort as a packet sniffer and develop their own IDS rules for this Lab.

Firewall is device or application that analyzes packet headers and enforces policy based on protocol type, source address, destination address, source port, and destination port. Packets that do not match policy are rejected.



Requirements

- VMware
- Ubuntu Linux (you can use any of your old ubuntu vm running)
- Snort
- Pfsense

Lab breakdown

Part 1: Install and configure snort

Part 2: Install pfsense and install snort on pfsense

Part 3: Answer questions

Part 1

Install snort on an ubuntu machine. Document your installation including showing a successful running SNORT install.

You can find more information on <https://www.snort.org/#get-started>



Alternative way of installing Snort (easier way/preferred way)

1. Installing snort from source can be very difficult. Run these two commands to make your life easy:

```
sudo apt-get -y update
```

```
sudo apt -y install snort
```

(Attach at least 2 screenshots)

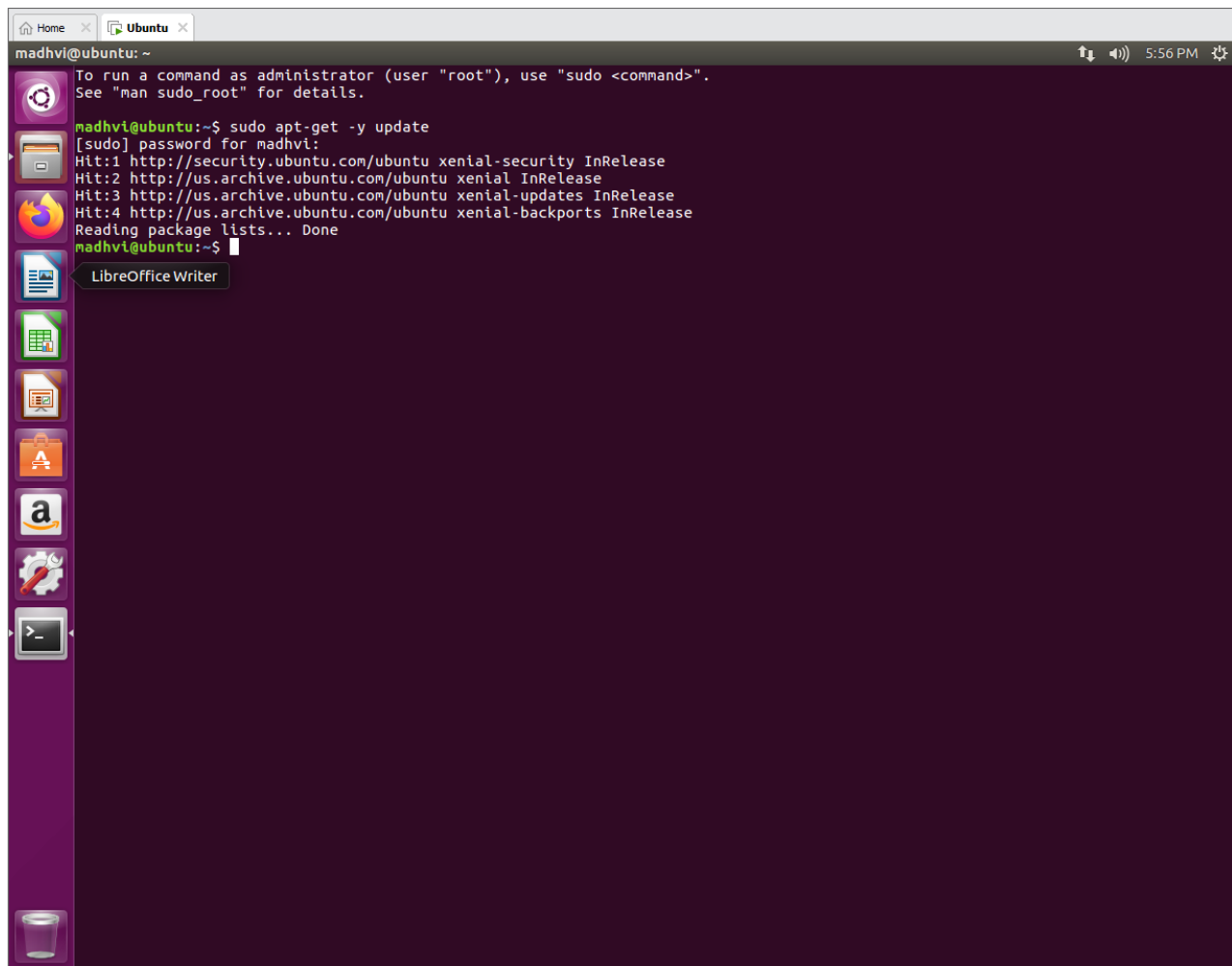
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

```
madhvi@ubuntu:~$ sudo apt-get -y update
[sudo] password for madhvi:
Hit:1 http://security.ubuntu.com/ubuntu xenial-security InRelease
Hit:2 http://us.archive.ubuntu.com/ubuntu xenial InRelease
Hit:3 http://us.archive.ubuntu.com/ubuntu xenial-updates InRelease
Hit:4 http://us.archive.ubuntu.com/ubuntu xenial-backports InRelease
Reading package lists... Done
madhvi@ubuntu:~$
```

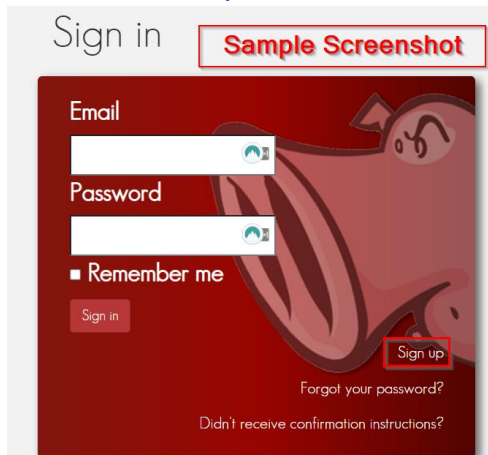
Lab 6: Firewall & Intrusion Detection Systems (Individual Lab)

INFT 1202

Total Marks :20



2. Set up a snort account on snort website, this would make it easy to download snort rules. Link: https://www.snort.org/users/sign_in

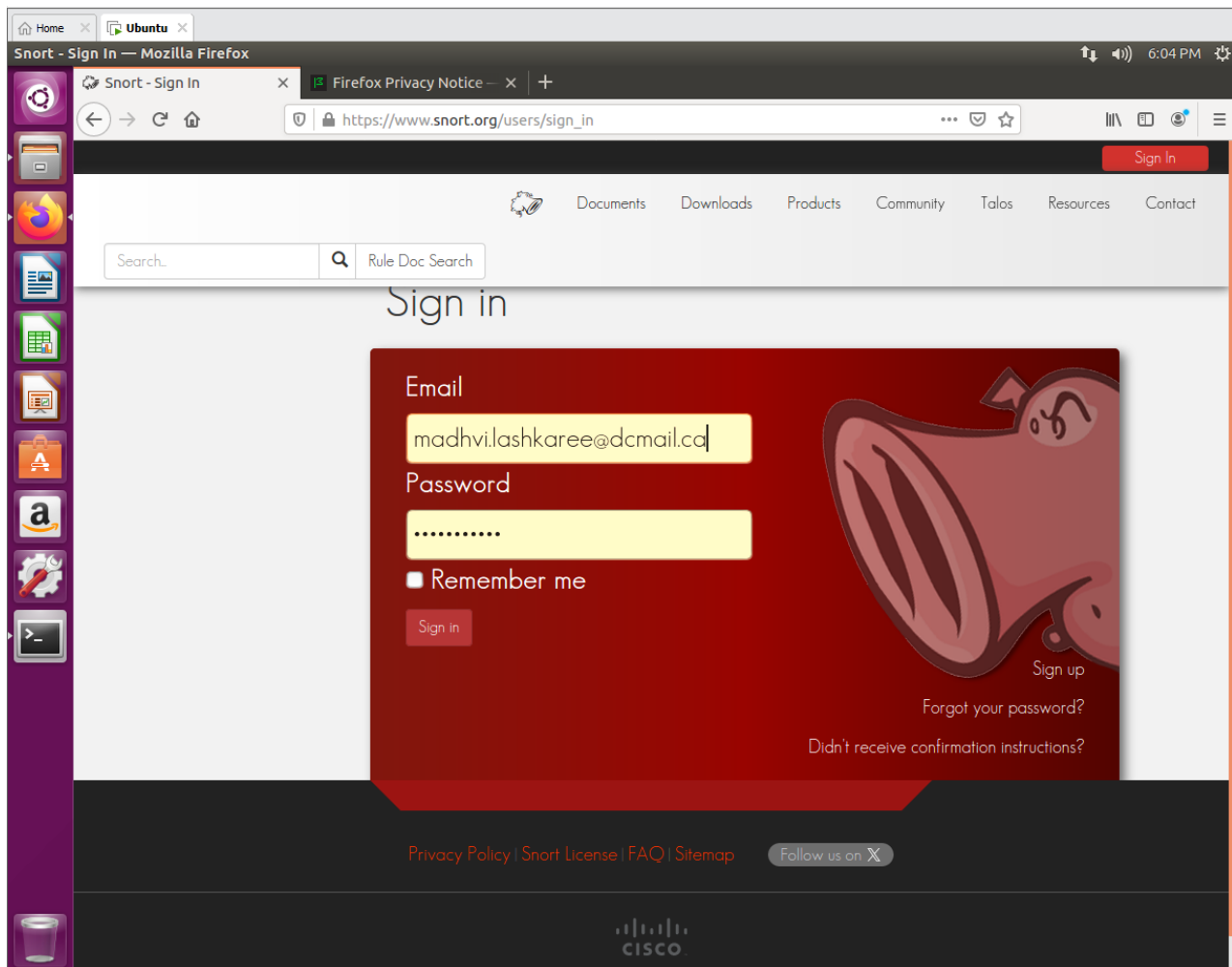


Lab 6: Firewall & Intrusion Detection Systems (Individual Lab)

INFT 1202

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Attach your screenshot



Set up snort

After installation open configuration files

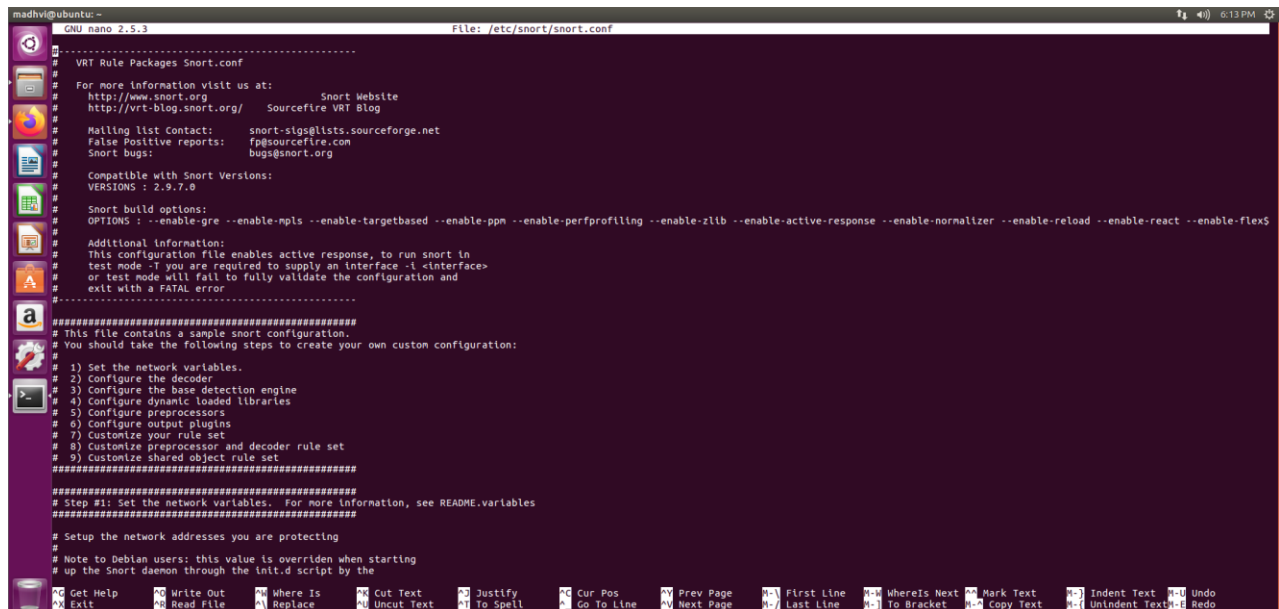
1. The configuration file of snort is stored at `/etc/snort/snort.conf`

Use nano, gedit or vim to edit the file.

(Note, switch to root user and you need to install vim as its not installed by default), Editing this configuration file is optional, but open it and take your screenshot.

Command: `nano /etc/snort/snort.conf`

Attach your screenshot



```
madhvi@ubuntu:~$ nano /etc/snort/snort.conf
GNU nano 2.5.3 File: /etc/snort/snort.conf
-----
# VRT Rule Packages Snort.conf
#
# For more information visit us at:
#   http://www.snort.org/      Sourcefire VRT Website
#   http://vrt-blog.snort.org/ Sourcefire VRT Blog
#
# Mailing list Contact:  snort-sigs@lists.sourceforge.net
# False Positive reports: fp@sourcefire.com
# Snort bugs:           bugs@snort.org
#
# Compatible with Snort Versions:
# VERSIONS : 2.9.7.0
#
# Snort build options:
# OPTIONS : --enable-gre --enable-mpls --enable-targetbased --enable-ppm --enable-perfprofiling --enable-zlib --enable-active-response --enable-normalizer --enable-reload --enable-react --enable-flex5
#
# Additional Information:
# This configuration file enables active response, to run snort in
# test mode -t you are required to supply an interface -i <interface>
# or test mode will fail to fully validate the configuration and
# exit with a FATAL error
#
# =====
# This file contains a sample snort configuration.
# You should take the following steps to create your own custom configuration:
#
# 1) Set the network variables.
# 2) Configure the decoder
# 3) Configure the base detection engine
# 4) Configure dynamic loaded libraries
# 5) Configure preprocessors
# 6) Configure output plugins
# 7) Customize your rule set
# 8) Customize preprocessor and decoder rule set
# 9) Customize shared object rule set
# =====
# Step #1: Set the network variables.  For more information, see README.variables
# =====
# Setup the network addresses you are protecting
#
# Note to Debian users: this value is overridden when starting
# up the Snort daemon through the init.d script by the
```

2. After installing the Snort, check if snort is running. run the command to check if snort service is running `service snort status`. If service is not running, use the below command to start service

`$ service snort start`

or

`$ /etc/init.d/snort start`

Lab 6: Firewall & Intrusion Detection Systems (Individual Lab)

INFT 1202

Total Marks :20

Attach your screenshot

```
madhvi@ubuntu:~$ sudo service snort status
● snort.service - LSB: Lightweight network intrusion detection system
   Loaded: loaded (/etc/init.d/snort; bad; vendor preset: enabled)
   Active: active (running) since Mon 2025-03-31 19:05:06 PDT; 11s ago
     Docs: man:systemd-sysv-generator(8)
  Process: 5102 ExecStart=/etc/init.d/snort start (code=exited, status=0/SUCCESS)
    CGroup: /system.slice/snort.service
            └─5116 /usr/sbin/snort -m 027 -D -d -l /var/log/snort -u snort -g snort -c /etc/snort/

Mar 31 19:05:06 ubuntu snort[5116]:      Preprocessor Object: SF_MODBUS Version 1.1 <Build
Mar 31 19:05:06 ubuntu snort[5116]:      Preprocessor Object: SF_SSH Version 1.1 <Build 3
Mar 31 19:05:06 ubuntu snort[5116]:      Preprocessor Object: SF_DCERPC2 Version 1.0 <Build
Mar 31 19:05:06 ubuntu snort[5116]:      Preprocessor Object: SF_FTPTELNET Version 1.2 <Build
Mar 31 19:05:06 ubuntu snort[5116]:      Preprocessor Object: SF_DNS Version 1.1 <Build 4
Mar 31 19:05:06 ubuntu snort[5116]:      Preprocessor Object: SF_IMAP Version 1.0 <Build
Mar 31 19:05:06 ubuntu snort[5116]:      Preprocessor Object: SF_SDF Version 1.1 <Build 1
Mar 31 19:05:06 ubuntu snort[5116]:      Preprocessor Object: SF_SSLPP Version 1.1 <Build
Mar 31 19:05:06 ubuntu snort[5116]:      Preprocessor Object: SF_SMTP Version 1.1 <Build
Mar 31 19:05:06 ubuntu snort[5116]: Commencing packet processing (pid=5116)
lines 1-18/18 (END)
```

Snort Rules

Snort is a signature-based IDS, and it defines rules to detect the intrusions. All rules of Snort are stored under `/etc/snort/rules` directory.

1. Run the command `ls /etc/snort/rules`

Attach your screenshot



```
madhvi@ubuntu:~$ ls /etc/snort/rules
attack-responses.rules      community-web-dos.rules    policy.rules
backdoor.rules             community-web-iis.rules    pop2.rules
bad-traffic.rules          community-web-misc.rules   pop3.rules
chat.rules                 community-web-php.rules    porn.rules
community-bot.rules        ddos.rules                rpc.rules
community-deleted.rules    deleted.rules             rservices.rules
community-dos.rules        dns.rules                scan.rules
community-exploit.rules    dos.rules                shellcode.rules
community-ftp.rules        experimental.rules        smtp.rules
community-game.rules       exploit.rules             snmp.rules
community-icmp.rules       finger.rules             sql.rules
community-imap.rules       ftp.rules                telnet.rules
community-inappropriate.rules icmp-info.rules          tftp.rules
community-mail-client.rules icmp.rules                virus.rules
community-misc.rules       imap.rules               web-attacks.rules
community-nntp.rules       info.rules               web-cgi.rules
community-oracle.rules     local.rules              web-client.rules
community-policy.rules     misc.rules               web-coldfusion.rules
community-sip.rules        multimedia.rules          web-frontpage.rules
community-smtp.rules       mysql.rules              web-iis.rules
community-sql-injection.rules netbios.rules            web-misc.rules
community-virus.rules      nntp.rules              web-php.rules
community-web-attacks.rules oracle.rules              x11.rules
community-web-cgi.rules    other-ids.rules
community-web-client.rules p2p.rules
madhvi@ubuntu:~$
```

2. Run the command `apt install vim` to install vim on your vm

Lab 6: Firewall & Intrusion Detection Systems (Individual Lab)

INFT 1202

Total Marks :20

- Then run the command `vim /etc/snort/rules/web-misc.rules`. This would open one of the rules, you can open any rule of your choice.

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```
## Copyright 2001-2005 Sourcefire, Inc. All Rights Reserved
##
## This file may contain proprietary rules that were created, tested and
## certified by Sourcefire, Inc. (the "VRT Certified Rules") as well as
## rules that were created by Sourcefire and other third parties and
## distributed under the GNU General Public License (the "GPL Rules"). The
## VRT Certified Rules contained in this file are the property of
## Sourcefire, Inc. Copyright 2005 Sourcefire, Inc. All Rights Reserved.
## The GPL Rules created by Sourcefire, Inc. are the property of
## Sourcefire, Inc. Copyright 2002-2005 Sourcefire, Inc. All Rights
## Reserved. All other GPL Rules are owned and copyrighted by their
## respective owners (please see www.snort.org/contributors for a list of
## owners and their respective copyrights). In order to determine what
## rules are VRT Certified Rules or GPL Rules, please refer to the VRT
## Certified Rules License Agreement.
##
##
## Sid: web-misc.rules.v 1.118.2.8.2.6 2005/07/22 19:19:54 mwatchinski Exp $
##-----
## WEB-MISC RULES
##-----

alert tcp $EXTERNAL_NET any -> $HTTP_SERVERS $HTTP_PORTS (msg:"WEB-MISC SSLV2 Client Hello with pad Challenge Length overflow attempt"; flow:to_server,established; flowbits:isnotset,sslv2.client_hello.request; flowbits:isnotset,sslv2.client_hello.request; byte_test:1,<128,0; content:"|01|"; depth:1; offset:3; byte_test:2,<4,768,4; flowbits:set,sslv2.client_hello.request; byte_test:3,>32,10; classtype:attempted-admin; sid:2659; rev:2)
alert tcp $EXTERNAL_NET any -> $HTTP_SERVERS $HTTP_PORTS (msg:"WEB-MISC SSLV2 Client Hello Challenge Length overflow attempt"; flow:to_server,established; flowbits:isnotset,sslv2.client_hello.request; flowbits:isnotset,sslv2.client_hello.request; flowbits:isnotset,sslv2.client_hello.request; flowbits:isnotset,sslv2.client_hello.request; byte_test:1,>127,0; content:"|01|"; depth:1; offset:2; byte_test:2,<4,768,3; flowbits:set,sslv2.client_hello.request; byte_test:3,>32,9; classtype:attempted-admin; sid:2658; rev:1)
alert tcp $EXTERNAL_NET any -> $HTTP_SERVERS $HTTP_PORTS (msg:"WEB-MISC cross site scripting attempt"; flow:to_server,established; content:"<SCRIPT>"; nocase; classtype:web-application-attack; sid:1497; rev:6)
alert tcp $EXTERNAL_NET any -> $HTTP_SERVERS $HTTP_PORTS (msg:"WEB-MISC cross site scripting HTML Image tag set to javascript attempt"; flow:to_server,established; content:"img src=javascript"; nocase; reference:bugtraq,4858; reference:cve,2002-0982; classtype:web-application-attack; sid:1667; rev:7)
alert tcp $EXTERNAL_NET any -> $HTTP_SERVERS $HTTP_PORTS (msg:"WEB-MISC Cisco IOS HTTP configuration attempt"; flow:to_server,established; uricontent:"/level/"; uricontent:"/exec/"; reference:bugtraq,2938; reference:cve,2001-0537; classtype:web-application-attack; sid:1250; rev:11)
alert tcp $EXTERNAL_NET any -> $HTTP_SERVERS $HTTP_PORTS (msg:"WEB-MISC Netscape Enterprise DOS"; flow:to_server,established; content:"REVL0G / "; depth:9; reference:bugtraq,2294; reference:cve,2001-0251; classtype:web-application-attack; sid:1047; rev:5)
alert tcp $EXTERNAL_NET any -> $HTTP_SERVERS $HTTP_PORTS (msg:"WEB-MISC Netscape Enterprise directory listing attempt"; flow:to_server,established; content:"INDEX "; depth:6; reference:bugtraq,2285; reference:cve,2001-0250; classtype:web-application-attack; sid:1048; rev:9)
alert tcp $EXTERNAL_NET any -> $HTTP_SERVERS $HTTP_PORTS (msg:"WEB-MISC IPlanet GETPROPERTIES attempt"; flow:to_server,established; content:"GETPROPERTIES"; depth:13; reference:bugtraq,2732; reference:cve,2001-0746; classtype:web-application-attack; sid:1050; rev:11)
alert tcp $EXTERNAL_NET any -> $HTTP_SERVERS $HTTP_PORTS (msg:"WEB-MISC Tomcat view source attempt"; flow:to_server,established; uricontent:"%252Ejsp"; reference:bugtraq,2527; reference:cve,2001-0590; classtype:web-application-attack; sid:1056; rev:8)
alert tcp $EXTERNAL_NET any -> $HTTP_SERVERS $HTTP_PORTS (msg:"WEB-MISC ftp attempt"; flow:to_server,established; content:"ftp.exe"; nocase; classtype:web-application-activity; sid:1057; rev:6)
alert tcp $EXTERNAL_NET any -> $HTTP_SERVERS $HTTP_PORTS (msg:"WEB-MISC xp_enundat attempt"; flow:to_server,established; content:"xp_enundat"; nocase; classtype:web-application-attack; sid:1058; rev:6)
alert tcp $EXTERNAL_NET any -> $HTTP_SERVERS $HTTP_PORTS (msg:"WEB-MISC xp_filelist attempt"; flow:to_server,established; content:"xp_filelist"; nocase; classtype:web-application-attack; sid:1059; rev:6)
alert tcp $EXTERNAL_NET any -> $HTTP_SERVERS $HTTP_PORTS (msg:"WEB-MISC xp_availability attempt"; flow:to_server,established; content:"xp_availability"; nocase; classtype:web-application-attack; sid:1060; rev:6)
alert tcp $EXTERNAL_NET any -> $HTTP_SERVERS $HTTP_PORTS (msg:"WEB-MISC xp_cndshell attempt"; flow:to_server,established; content:"xp_cndshell"; nocase; classtype:web-application-attack; sid:1061; rev:6)
alert tcp $EXTERNAL_NET any -> $HTTP_SERVERS $HTTP_PORTS (msg:"WEB-MISC nc.exe attempt"; flow:to_server,established; content:"nc.exe"; nocase; classtype:web-application-activity; sid:1062; rev:6)
alert tcp $EXTERNAL_NET any -> $HTTP_SERVERS $HTTP_PORTS (msg:"WEB-MISC wsh attempt"; flow:to_server,established; content:"wsh.exe"; nocase; classtype:web-application-activity; sid:1064; rev:6)
alert tcp $EXTERNAL_NET any -> $HTTP_SERVERS $HTTP_PORTS (msg:"WEB-MISC rcnd attempt"; flow:to_server,established; uricontent:"rcnd.exe"; nocase; classtype:web-application-activity; sid:1065; rev:7)
1,1 Top
```

Writing and Adding a Snort Rule

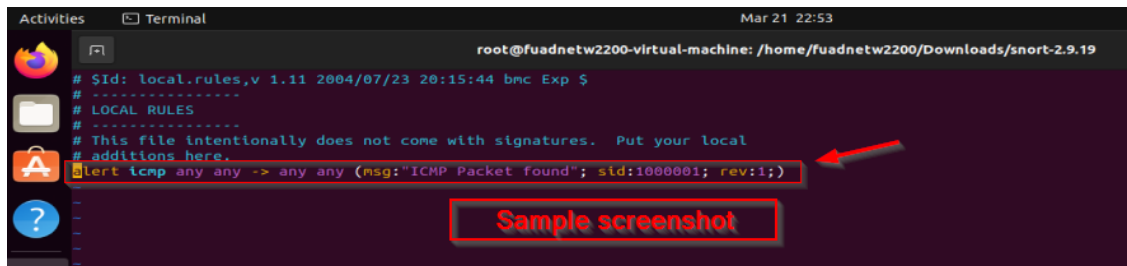
You should add your own rules at /etc/snort/rules/local.rules.

Run the command **vim /etc/snort/rules/local.rules** (you can use also use nano or gedit)

Note: for vim, type the **i key to insert on vim and press the escape key and type the command **:wq** to quit and save**

Add the following line into the local.rules file

alert icmp any any -> any any (msg:"ICMP Packet found"; sid:1000001; rev:1;)

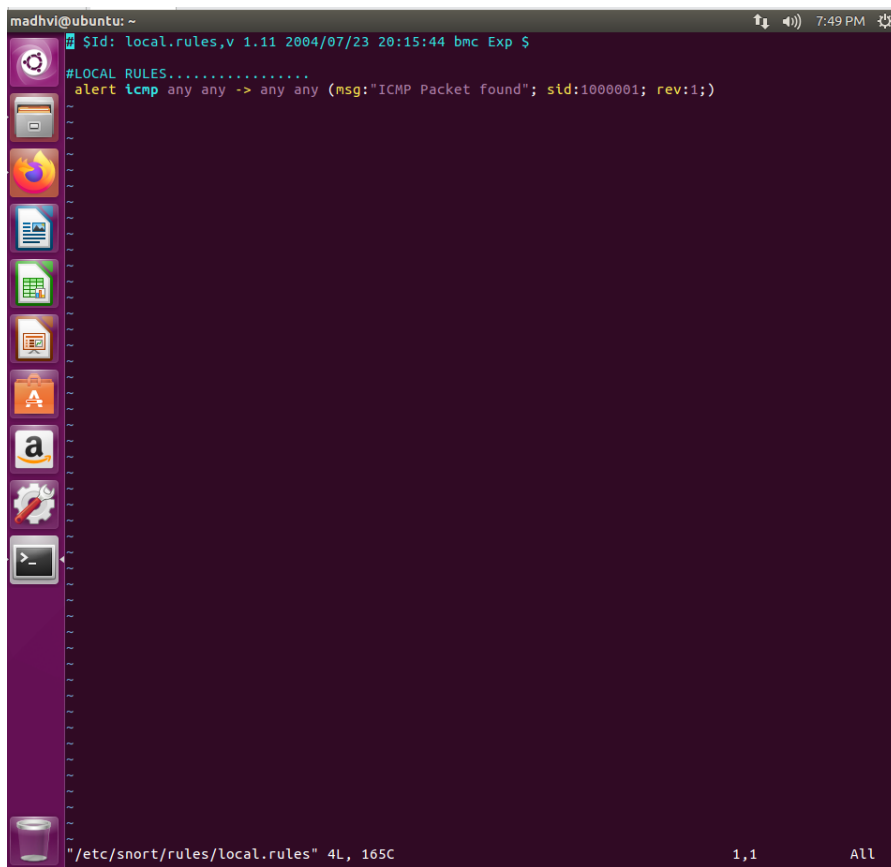


Lab 6: Firewall & Intrusion Detection Systems (Individual Lab)

INFT 1202

Total Marks :20

Attach your screenshot



```
madhvi@ubuntu: ~  
$Id: local.rules,v 1.11 2004/07/23 20:15:44 bmc Exp $  
#LOCAL RULES.....  
alert icmp any any -> any any (msg:"ICMP Packet found"; sid:1000001; rev:1;)
```

To make the rule become effective, you need to restart the snort service by typing the following command.

```
$ sudo service snort restart
```

or

```
$ sudo /etc/init.d/snort restart
```

Triggering an Alert for the New Rule

To trigger an alert for the new rule, you only need to send an ICMP message such as a ping to the VM image where snort is running.

1. First, you need to find the IP address of the VM snort is installed and confirm the IP address of your attacker machine (kali) by typing **Ifconfig**

```
fuad@ubuntu:~/Desktop$ ifconfig
ens33: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.80.226 netmask 255.255.255.0 broadcast 192.168.80.255
    inet6 fe80::af2e:ee20:88ef:e805 prefixlen 64 scopeid 0x20<link>
    ether 00:0c:29:3c:8c:f6 txqueuelen 1000 (Ethernet)
    RX packets 3024 bytes 3635280 (3.6 MB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 975 bytes 101151 (101.1 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```



```
[03/23/22] fuad@h:\w$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.80.181 netmask 255.255.255.0 broadcast 192.168.80.255
    ether 00:0c:29:07:da:4a txqueuelen 1000 (Ethernet)
    RX packets 626672 bytes 148531034 (141.6 MiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 1148854 bytes 100154401 (95.5 MiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.44.129 netmask 255.255.255.0 broadcast 192.168.44.255
    inet6 fe80::20c:29ff:fe07:da54 prefixlen 64 scopeid 0x20<link>
    ether 00:0c:29:07:da:54 txqueuelen 1000 (Ethernet)
    RX packets 3627 bytes 374785 (366.0 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 9632 bytes 937295 (915.3 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 276793 bytes 24374523 (23.2 MiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 276793 bytes 24374523 (23.2 MiB)
```



Lab 6: Firewall & Intrusion Detection Systems (Individual Lab)

INFT 1202

Total Marks :20

Attach your screenshot of Ubuntu machine IP

```
madhvi@ubuntu:~$ ifconfig
ens33      Link encap:Ethernet  HWaddr 00:0c:29:74:8b:65
            inet addr:192.168.81.141  Bcast:192.168.81.255  Mask:255.255.255.0
            inet6 addr: fe80::bc75:f64d:6301:4234/64 Scope:Link
            UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
            RX packets:28055 errors:62 dropped:0 overruns:0 frame:0
            TX packets:14881 errors:0 dropped:0 overruns:0 carrier:0
            collisions:0 txqueuelen:1000
            RX bytes:31397963 (31.3 MB)  TX bytes:1371823 (1.3 MB)
            Interrupt:19 Base address:0x2000

lo         Link encap:Local Loopback
            inet addr:127.0.0.1  Mask:255.0.0.0
            inet6 addr: ::1/128 Scope:Host
            UP LOOPBACK RUNNING  MTU:65536  Metric:1
            RX packets:3046 errors:0 dropped:0 overruns:0 frame:0
            TX packets:3046 errors:0 dropped:0 overruns:0 carrier:0
            collisions:0 txqueuelen:1000
            RX bytes:268687 (268.6 KB)  TX bytes:268687 (268.6 KB)

madhvi@ubuntu:~$
```

Attach your screenshot of kali machine IP

```
Madhvi$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.81.139  netmask 255.255.255.0  broadcast 192.168.81.255
        inet6 fe80::7c74:6940:8972:9101  prefixlen 64  scopeid 0x20<link>
        ether 00:0c:29:a1:66:e2  txqueuelen 1000  (Ethernet)
        RX packets 54  bytes 3522 (3.4 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 28  bytes 3400 (3.3 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop txqueuelen 1000  (Local Loopback)
        RX packets 8  bytes 480 (480.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 8  bytes 480 (480.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

Madhvi$
```

Lab 6: Firewall & Intrusion Detection Systems (Individual Lab)

INFT 1202

Total Marks :20

2. Open a terminal in your Kali machine and ping your ubuntu vm running snort or ping your ubuntu from any other vm in your lab

```
[03/23/22] fuad@h:\w$ ping 192.168.80.226
PING 192.168.80.226 (192.168.80.226) 56(84) bytes of data.
64 bytes from 192.168.80.226: icmp_seq=1 ttl=64 time=0.436 ms
64 bytes from 192.168.80.226: icmp_seq=2 ttl=64 time=1.07 ms
64 bytes from 192.168.80.226: icmp_seq=3 ttl=64 time=2.80 ms
64 bytes from 192.168.80.226: icmp_seq=4 ttl=64 time=0.452 ms
^C
--- 192.168.80.226 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3017ms
rtt min/avg/max/mdev = 0.436/1.187/2.795/0.962 ms

[03/23/22] fuad@h:\w$
```



Attach your screenshot

```
Madhvi$ ping 192.168.81.141
PING 192.168.81.141 (192.168.81.141) 56(84) bytes of data.
64 bytes from 192.168.81.141: icmp_seq=1 ttl=64 time=2.11 ms
64 bytes from 192.168.81.141: icmp_seq=2 ttl=64 time=1.53 ms
64 bytes from 192.168.81.141: icmp_seq=3 ttl=64 time=0.538 ms
64 bytes from 192.168.81.141: icmp_seq=4 ttl=64 time=1.69 ms
64 bytes from 192.168.81.141: icmp_seq=5 ttl=64 time=0.643 ms
64 bytes from 192.168.81.141: icmp_seq=6 ttl=64 time=1.42 ms
64 bytes from 192.168.81.141: icmp_seq=7 ttl=64 time=0.805 ms
64 bytes from 192.168.81.141: icmp_seq=8 ttl=64 time=0.785 ms
64 bytes from 192.168.81.141: icmp_seq=9 ttl=64 time=0.750 ms
64 bytes from 192.168.81.141: icmp_seq=10 ttl=64 time=1.71 ms
64 bytes from 192.168.81.141: icmp_seq=11 ttl=64 time=0.446 ms
```

3. After you send the ping messages, the alerts should be triggered, and you can find the log messages in /var/log/snort/snort.log. However, the snort.log file will be binary format. You need to use a tool, called u2spewfoo, to read it. The screenshot below shows the result of reading the snort alerts.

Command: u2spewfoo /var/log/snort/snort.log

Lab 6: Firewall & Intrusion Detection Systems (Individual Lab)

INFT 1202

Total Marks :20

```
Read 25266 of 892219957 bytes
root@fuadnetw2200-virtual-machine: /home/fuadnetw2200/Downloads/snort-2.9.19# u2spewfoo /var/log/snort/snort.log

(IPv6 Event)
  sensor id: 0      event id: 1      event second: 1647928757      event microsecond: 40899
  sig id: 1000001  gen id: 1      revision: 1      classification: 0
  priority: 0      ip source: ::      ip destination: ff02::16
  src port: 0      dest port: 0      protocol: 58      impact_flag: 0      blocked: 0
  mpls label: 0    vland id: 0      policy id: 0      appid:

Packet
  sensor id: 0      event id: 1      event second: 1647928757      packet microsecond: 40899
  packet second: 1647928757      packet length: 110
  linktype: 1
[ 0] 33 33 00 00 00 10 00 0C 29 58 98 96 86 D0 60 00 33.....)X....'.
[ 16] 00 00 00 38 00 01 00 00 00 00 00 00 00 00 00 00 ...8.....
[ 32] 00 00 00 00 00 00 FF 02 00 00 00 00 00 00 00 00 .....
[ 48] 00 00 00 00 00 16 3A 00 05 02 00 00 01 00 8F 00 .....:.....
[ 64] 92 0A 00 00 00 02 04 00 00 00 FF 02 00 00 00 00 .....
[ 80] 00 00 00 00 00 00 00 00 00 00 FB 04 00 00 00 FF 02 .....
[ 96] 00 00 00 00 00 00 00 00 01 FF 2C D9 40 .....@

(IPv6 Event)
  sensor id: 0      event id: 2      event second: 1647928757      event microsecond: 40899
  sig id: 527      gen id: 1      revision: 8      classification: 3
  priority: 2      ip source: ::      ip destination: ff02::16
  src port: 0      dest port: 0      protocol: 58      impact_flag: 0      blocked: 0
  mpls label: 0    vland id: 0      policy id: 0      appid:

Packet
  sensor id: 0      event id: 2      event second: 1647928757      packet microsecond: 40899
  packet second: 1647928757      packet length: 110
  linktype: 1
[ 0] 33 33 00 00 00 10 00 0C 29 58 98 96 86 D0 60 00 33.....)X....'.
[ 16] 00 00 00 38 00 01 00 00 00 00 00 00 00 00 00 00 ...8.....
[ 32] 00 00 00 00 00 00 FF 02 00 00 00 00 00 00 00 00 .....
[ 48] 00 00 00 00 00 16 3A 00 05 02 00 00 01 00 8F 00 .....:.....
[ 64] 92 0A 00 00 00 02 04 00 00 00 FF 02 00 00 00 00 .....
[ 80] 00 00 00 00 00 00 00 00 00 00 FB 04 00 00 00 FF 02 .....
[ 96] 00 00 00 00 00 00 00 00 01 FF 2C D9 40 .....@

(Event)
  sensor id: 0      event id: 3      event second: 1647928757      event microsecond: 46862
  sig id: 527      gen id: 1      revision: 8      classification: 3
  priority: 2      ip source: 0.0.0.0      ip destination: 255.255.255.255
```

Sample Screenshot

You can see that the SID is 1000001, and the alerts are generated by the ICMP messages.

Lab 6: Firewall & Intrusion Detection Systems (Individual Lab)

INFT 1202

Total Marks :20

Attach your screenshot

```
madhvi@ubuntu: ~
mpls label: 0  vland id: 0  policy id: 0

Packet
sensor id: 0  event id: 1  event second: 1743476105
packet second: 1743476105  packet microsecond: 344130
linktype: 1  packet_length: 330
[ 0] FF FF FF FF FF 00 0C 29 A1 66 E2 08 00 45 00 .....).f...E.
[16] 01 3C 00 00 40 00 40 11 39 B2 00 00 00 00 FF FF <...@.@.9.....
[32] FF FF 00 44 00 43 01 28 7B AF 01 01 06 00 3B 8C ...D.C.({.....;
[48] 12 C3 00 01 00 00 00 00 00 00 00 00 00 00 00 00 .....
[64] 00 00 00 00 00 00 00 0C 29 A1 66 E2 00 00 00 00 .....).f.....
[80] 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
[96] 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
[112] 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
[128] 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
[144] 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
[160] 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
[176] 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
[192] 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
[208] 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
[224] 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
[240] 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
[256] 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
[272] 00 00 00 00 00 00 63 82 53 63 35 01 03 3D 07 01 .....c.Sc5..=.
[288] 00 0C 29 A1 66 E2 37 11 01 02 06 0C 0F 1A 1C 79 ..).f.7.....y
[304] 03 21 28 29 2A 77 F9 FC 11 39 02 02 40 32 04 C0 .!()*w...9..@2..
[320] A8 51 8B 0C 04 6B 61 6C 69 FF .Q...kali.

(IPv6 Event)
sensor id: 0  event id: 2  event second: 1743476105  event microsecond: 352334
sig id: 1000001  gen id: 1  revision: 1  classification: 0
priority: 0  ip source: ::  ip destination: ff02::1:ff72:9101
src port: 0  dest port: 0  protocol: 58  impact_flag: 0  blocked: 0
mpls label: 0  vland id: 0  policy id: 0

Packet
sensor id: 0  event id: 2  event second: 1743476105
packet second: 1743476105  packet microsecond: 352334
linktype: 1  packet_length: 86
[ 0] 33 33 FF 72 91 01 00 0C 29 A1 66 E2 86 DD 60 00 33.r....).f...`.
[16] 00 00 00 20 3A FF 00 00 00 00 00 00 00 00 00 00 ... :.....
[32] 00 00 00 00 00 00 FF 02 00 00 00 00 00 00 00 00 .....
[48] 00 01 FF 72 91 01 87 00 32 2F 00 00 00 00 FE 80 ...f....2/.....
[64] 00 00 00 00 00 00 7C 74 69 40 89 72 91 01 0E 01 .....|tl@.r....
[80] 1F 5F 24 E3 66 10 ._S.f.

(IPv6 Event)
sensor id: 0  event id: 3  event second: 1743476105  event microsecond: 352334
sig id: 527  gen id: 1  revision: 8  classification: 3
priority: 2  ip source: ::  ip destination: ff02::1:ff72:9101
src port: 0  dest port: 0  protocol: 58  impact_flag: 0  blocked: 0
mpls label: 0  vland id: 0  policy id: 0
```

4. Next step would be to listen in real time. We would run snort to sniff the traffic in real time. We would run the command below to sniff traffic

Use the command “**snort -A console -c /etc/snort/snort.conf**”

```
root@ubuntu:/etc/snort# snort -A console -c /etc/snort/snort.conf
```


Lab 6: Firewall & Intrusion Detection Systems (Individual Lab)

INFT 1202

Total Marks :20

```
Preprocessor Object: SF_SMTP Version 1.1 <Build 9>
Preprocessor Object: apid Version 1.1 <Build 5>
Preprocessor Object: SF_MODBUS Version 1.1 <Build 1>
Preprocessor Object: SF_SSLPP Version 1.1 <Build 4>
Preprocessor Object: SF_SIP Version 1.1 <Build 1>
Preprocessor Object: SF_FTPTELNET Version 1.2 <Build 13>
Commencing packet processing (pid=6349)
03/23-19:25:51.683466 1:1000001:1 ICMP Packet found [Priority: 0] [ICMP] 192.168.80.181 -> 192.168.80.226
03/23-19:25:51.683498 1:1000001:1 ICMP Packet found [Priority: 0] [ICMP] 192.168.80.226 -> 192.168.80.181
03/23-19:25:52.684175 1:1000001:1 ICMP Packet found [Priority: 0] [ICMP] 192.168.80.181 -> 192.168.80.226
03/23-19:25:52.684230 1:1000001:1 ICMP Packet found [Priority: 0] [ICMP] 192.168.80.226 -> 192.168.80.181
03/23-19:25:53.699786 1:1000001:1 ICMP Packet found [Priority: 0] [ICMP] 192.168.80.181 -> 192.168.80.226
03/23-19:25:53.699849 1:1000001:1 ICMP Packet found [Priority: 0] [ICMP] 192.168.80.226 -> 192.168.80.181
03/23-19:25:54.700856 1:1000001:1 ICMP Packet found [Priority: 0] [ICMP] 192.168.80.181 -> 192.168.80.226
03/23-19:25:54.700891 1:1000001:1 ICMP Packet found [Priority: 0] [ICMP] 192.168.80.226 -> 192.168.80.181
```

Kali IP Address (Attacker Machine) → 192.168.80.181
Ubuntu IP Address of Snort machine → 192.168.80.226

Sample Screenshot

Attach your screenshot

```
Acquiring network traffic from "ens33".
Reload thread starting...
Reload thread started, thread 0x9eb2db04 (6758)
Decoding Ethernet
==== Initialization Complete ====

--> Snort! <--
Version 2.9.7.0 GRE (Build 149)
By Martin Roesch & The Snort Team: http://www.snort.org/contact#team
Copyright (C) 2014 Cisco and/or its affiliates. All rights reserved.
Copyright (C) 1998-2013 Sourcefire, Inc., et al.
Using libpcap version 1.7.4
Using PCRE version: 8.38 2015-11-23
Using ZLIB version: 1.2.8

Rules Engine: SF_SNORT DETECTION ENGINE Version 2.4 <Build 1>
Preprocessor Object: SF_SIP Version 1.1 <Build 1>
Preprocessor Object: SF_DNP3 Version 1.1 <Build 1>
Preprocessor Object: SF_POP Version 1.0 <Build 1>
Preprocessor Object: SF_REPUTATION Version 1.1 <Build 1>
Preprocessor Object: SF_GTP Version 1.1 <Build 1>
Preprocessor Object: SF_MODBUS Version 1.1 <Build 1>
Preprocessor Object: SF_SSH Version 1.1 <Build 3>
Preprocessor Object: SF_DCEAPC2 Version 1.0 <Build 3>
Preprocessor Object: SF_FTPTELNET Version 1.2 <Build 13>
Preprocessor Object: SF_DNS Version 1.1 <Build 4>
Preprocessor Object: SF_INAP Version 1.0 <Build 1>
Preprocessor Object: SF_SOF Version 1.1 <Build 1>
Preprocessor Object: SF_SSLPP Version 1.1 <Build 4>
Preprocessor Object: SF_SMTP Version 1.1 <Build 9>
Commencing packet processing (pid=6753)
03/31-20:18:04.230111 1:1366:7 ICMP PING *NIX [Classification: Misc activity] [Priority: 3] [ICMP] 192.168.81.139 -> 192.168.81.141
03/31-20:18:04.230111 1:1384:5 ICMP PING [Classification: Misc activity] [Priority: 3] [ICMP] 192.168.81.139 -> 192.168.81.141
03/31-20:18:04.230131 1:1000001:1 ICMP Packet found [Priority: 0] [ICMP] 192.168.81.141 -> 192.168.81.139
03/31-20:18:04.230131 1:1408:5 ICMP Echo Reply [Classification: Misc activity] [Priority: 3] [ICMP] 192.168.81.141 -> 192.168.81.139
03/31-20:18:05.231434 1:1366:7 ICMP PING *NIX [Classification: Misc activity] [Priority: 3] [ICMP] 192.168.81.139 -> 192.168.81.141
03/31-20:18:05.231434 1:1384:5 ICMP PING [Classification: Misc activity] [Priority: 3] [ICMP] 192.168.81.139 -> 192.168.81.141
03/31-20:18:05.231434 1:1000001:1 ICMP Packet found [Priority: 0] [ICMP] 192.168.81.139 -> 192.168.81.141
03/31-20:18:05.231434 1:1408:5 ICMP Echo Reply [Classification: Misc activity] [Priority: 3] [ICMP] 192.168.81.141 -> 192.168.81.139
03/31-20:18:05.231464 1:1366:7 ICMP PING *NIX [Classification: Misc activity] [Priority: 3] [ICMP] 192.168.81.139 -> 192.168.81.141
03/31-20:18:05.231464 1:1384:5 ICMP PING [Classification: Misc activity] [Priority: 3] [ICMP] 192.168.81.139 -> 192.168.81.141
03/31-20:18:05.231464 1:1000001:1 ICMP Packet found [Priority: 0] [ICMP] 192.168.81.139 -> 192.168.81.141
03/31-20:18:05.231464 1:1408:5 ICMP Echo Reply [Classification: Misc activity] [Priority: 3] [ICMP] 192.168.81.141 -> 192.168.81.139
03/31-20:18:06.257206 1:1366:7 ICMP PING *NIX [Classification: Misc activity] [Priority: 3] [ICMP] 192.168.81.139 -> 192.168.81.141
03/31-20:18:06.257206 1:1384:5 ICMP PING [Classification: Misc activity] [Priority: 3] [ICMP] 192.168.81.139 -> 192.168.81.141
03/31-20:18:06.257206 1:1000001:1 ICMP Packet found [Priority: 0] [ICMP] 192.168.81.139 -> 192.168.81.141
03/31-20:18:06.257273 1:1408:5 ICMP Echo Reply [Classification: Misc activity] [Priority: 3] [ICMP] 192.168.81.141 -> 192.168.81.139
03/31-20:18:06.257273 1:1366:7 ICMP PING *NIX [Classification: Misc activity] [Priority: 3] [ICMP] 192.168.81.139 -> 192.168.81.141
03/31-20:18:06.257273 1:1384:5 ICMP PING [Classification: Misc activity] [Priority: 3] [ICMP] 192.168.81.139 -> 192.168.81.141
03/31-20:18:06.257273 1:1000001:1 ICMP Packet found [Priority: 0] [ICMP] 192.168.81.139 -> 192.168.81.141
03/31-20:18:06.257273 1:1408:5 ICMP Echo Reply [Classification: Misc activity] [Priority: 3] [ICMP] 192.168.81.141 -> 192.168.81.139
03/31-20:18:07.260527 1:1366:7 ICMP PING *NIX [Classification: Misc activity] [Priority: 3] [ICMP] 192.168.81.139 -> 192.168.81.141
03/31-20:18:07.260527 1:1384:5 ICMP PING [Classification: Misc activity] [Priority: 3] [ICMP] 192.168.81.139 -> 192.168.81.141
03/31-20:18:07.260527 1:1000001:1 ICMP Packet found [Priority: 0] [ICMP] 192.168.81.139 -> 192.168.81.141
03/31-20:18:07.260527 1:1408:5 ICMP Echo Reply [Classification: Misc activity] [Priority: 3] [ICMP] 192.168.81.141 -> 192.168.81.139
```

5. Use ctrl + C to end your sniffing and run it again, this time you will run an nmap scan against your target machine. Use the nmap command **nmap -sC -sV [UbuntuIP]**. Once you run this, wait for some time for snort to display the logs.

```
[03/23/22] fuada@h:\w$ nmap -sC -sV 192.168.80.226
Starting Nmap 7.92 ( https://nmap.org ) at 2022-03-23 23:17 EDT
Nmap scan report for 192.168.80.226
Host is up (0.00086s latency).
All 1000 scanned ports on 192.168.80.226 are in ignored states.
Not shown: 1000 closed tcp ports (reset)
MAC Address: 00:0C:29:3C:8C:F6 (VMware)

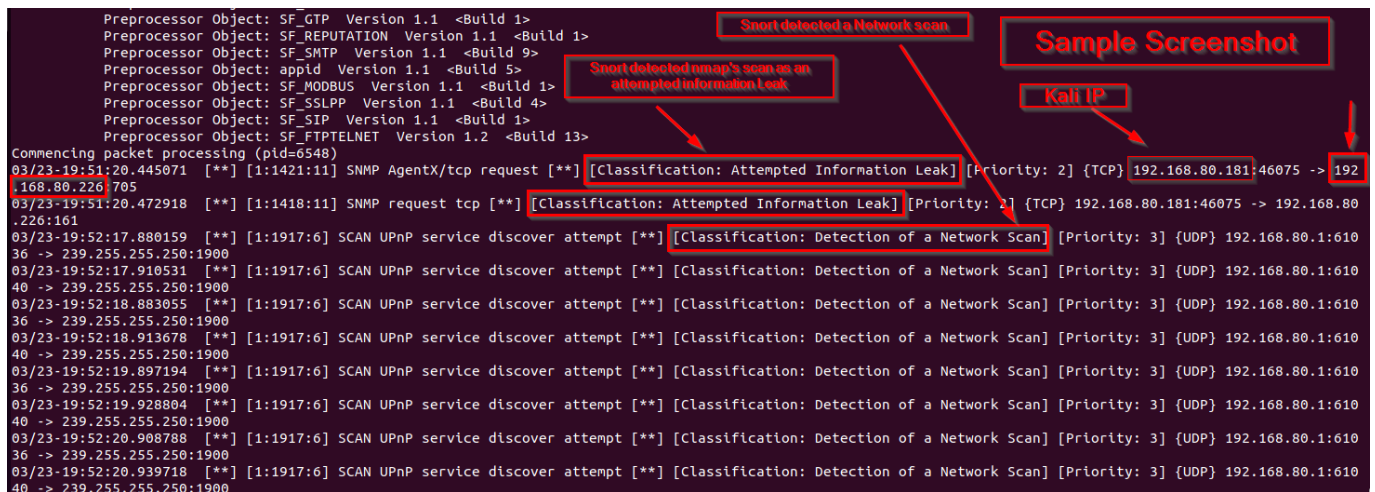
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 0.97 seconds
```

Sample Screenshot

Lab 6: Firewall & Intrusion Detection Systems (Individual Lab)

INFT 1202

Total Marks :20



Attach your screenshot of nmap scan

```
Madhvi$ nmap -sC -sV 192.168.81.141
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-03-31 23:19 EDT
Nmap scan report for 192.168.81.141
Host is up (0.0014s latency).
All 1000 scanned ports on 192.168.81.141 are in ignored states.
Not shown: 1000 closed tcp ports (reset)
MAC Address: 00:0C:29:74:8B:65 (VMware)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 1.62 seconds
```

Lab 6: Firewall & Intrusion Detection Systems (Individual Lab)

INFT 1202

Total Marks :20

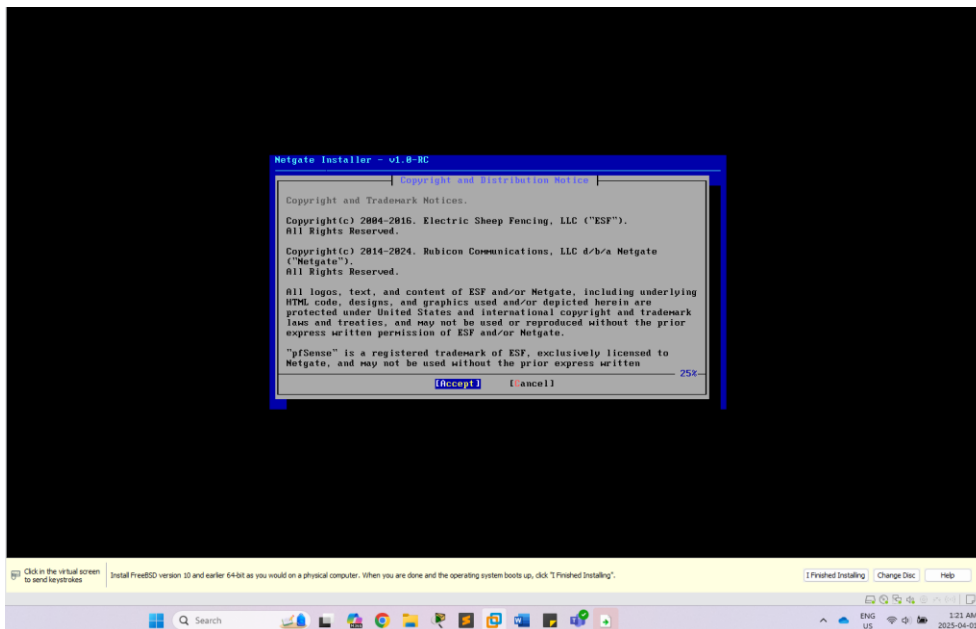
Attach your screenshot of nmap scan detected

```
03/31-20:18:22.352077 ** [1:408:5] ICMP Echo Reply ** [Classification: Misc activity] [Priority: 3] {ICMP} 192.168.81.141 -> 192.168.81.139
03/31-20:18:23.353770 ** [1:366:7] ICMP PING *NIX ** [Classification: Misc activity] [Priority: 3] {ICMP} 192.168.81.139 -> 192.168.81.141
03/31-20:18:23.353770 ** [1:1000001:1] ICMP Packet found ** [Priority: 0] {ICMP} 192.168.81.139 -> 192.168.81.141
03/31-20:18:23.353770 ** [1:384:5] ICMP PING ** [Classification: Misc activity] [Priority: 3] {ICMP} 192.168.81.139 -> 192.168.81.141
03/31-20:18:23.353812 ** [1:1000001:1] ICMP Packet found ** [Priority: 0] {ICMP} 192.168.81.141 -> 192.168.81.139
03/31-20:18:23.353812 ** [1:408:5] ICMP Echo Reply ** [Classification: Misc activity] [Priority: 3] {ICMP} 192.168.81.141 -> 192.168.81.139
03/31-20:18:24.355058 ** [1:366:7] ICMP PING *NIX ** [Classification: Misc activity] [Priority: 3] {ICMP} 192.168.81.139 -> 192.168.81.141
03/31-20:18:24.355058 ** [1:1000001:1] ICMP Packet found ** [Priority: 0] {ICMP} 192.168.81.139 -> 192.168.81.141
03/31-20:18:24.355058 ** [1:384:5] ICMP PING ** [Classification: Misc activity] [Priority: 3] {ICMP} 192.168.81.139 -> 192.168.81.141
03/31-20:18:24.355093 ** [1:1000001:1] ICMP Packet found ** [Priority: 0] {ICMP} 192.168.81.141 -> 192.168.81.139
03/31-20:18:24.355093 ** [1:408:5] ICMP Echo Reply ** [Classification: Misc activity] [Priority: 3] {ICMP} 192.168.81.141 -> 192.168.81.139
03/31-20:18:25.359378 ** [1:366:7] ICMP PING *NIX ** [Classification: Misc activity] [Priority: 3] {ICMP} 192.168.81.139 -> 192.168.81.141
03/31-20:18:25.359378 ** [1:1000001:1] ICMP Packet found ** [Priority: 0] {ICMP} 192.168.81.139 -> 192.168.81.141
03/31-20:18:25.359378 ** [1:384:5] ICMP PING ** [Classification: Misc activity] [Priority: 3] {ICMP} 192.168.81.139 -> 192.168.81.141
03/31-20:18:25.359420 ** [1:1000001:1] ICMP Packet found ** [Priority: 0] {ICMP} 192.168.81.141 -> 192.168.81.139
03/31-20:18:25.359420 ** [1:408:5] ICMP Echo Reply ** [Classification: Misc activity] [Priority: 3] {ICMP} 192.168.81.141 -> 192.168.81.139
03/31-20:18:52.159735 ** [1:1000001:1] ICMP Packet found ** [Priority: 0] {IPV6-ICMP} fe80::873:c2ab:c3c6:c3ab -> ff02::16
03/31-20:18:52.233207 ** [1:1000001:1] ICMP Packet found ** [Priority: 0] {IPV6-ICMP} fe80::873:c2ab:c3c6:c3ab -> ff02::16
03/31-20:18:52.234191 ** [1:1000001:1] ICMP Packet found ** [Priority: 0] {IPV6-ICMP} fe80::873:c2ab:c3c6:c3ab -> ff02::16
03/31-20:18:52.252467 ** [1:1000001:1] ICMP Packet found ** [Priority: 0] {IPV6-ICMP} fe80::873:c2ab:c3c6:c3ab -> ff02::16
03/31-20:18:52.455467 ** [1:1000001:1] ICMP Packet found ** [Priority: 0] {IPV6-ICMP} fe80::873:c2ab:c3c6:c3ab -> ff02::16
03/31-20:19:58.462724 ** [1:1418:11] SNMP request tcp ** [Classification: Attempted Information Leak] [Priority: 2] {TCP} 192.168.81.139:49442 -> 192.168.81.141:161
03/31-20:19:58.508192 ** [1:1421:11] SNMP AgentX/tcp request ** [Classification: Attempted Information Leak] [Priority: 2] {TCP} 192.168.81.139:49442 -> 192.168.81.141:705
03/31-20:19:59.696071 ** [1:1000001:1] ICMP Packet found ** [Priority: 0] {ICMP} 192.168.81.141 -> 192.168.81.139
03/31-20:19:59.696071 ** [1:402:7] ICMP Destination Unreachable Port Unreachable ** [Classification: Misc activity] [Priority: 3] {ICMP} 192.168.81.141 -> 192.168.81.139
```

Part 2

Install Pfsense and install snort module on pfsense

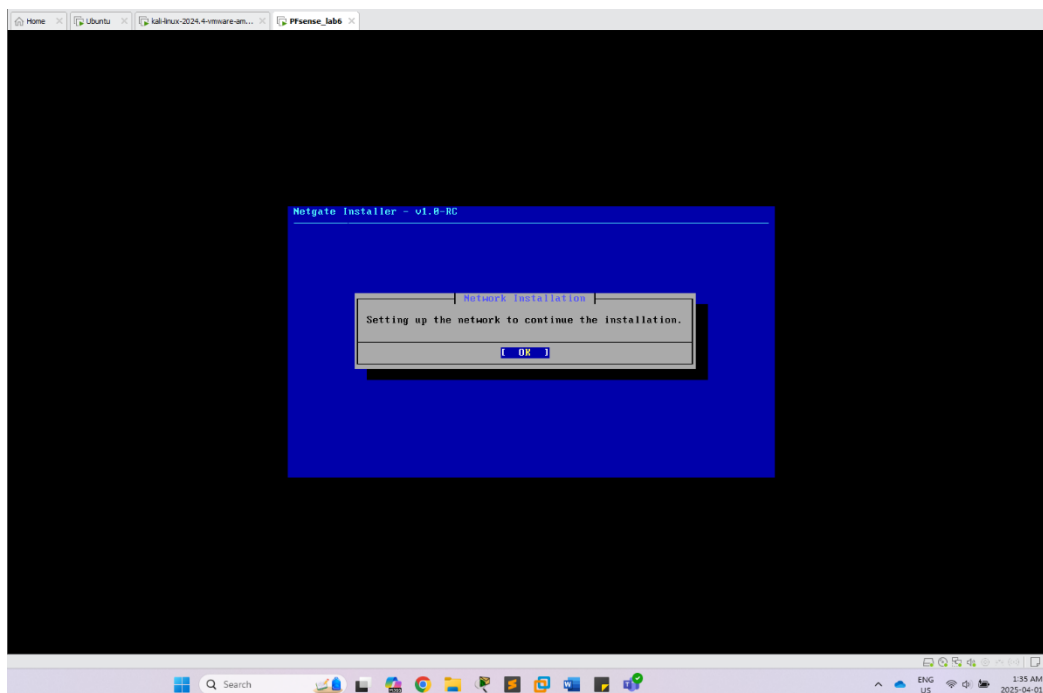
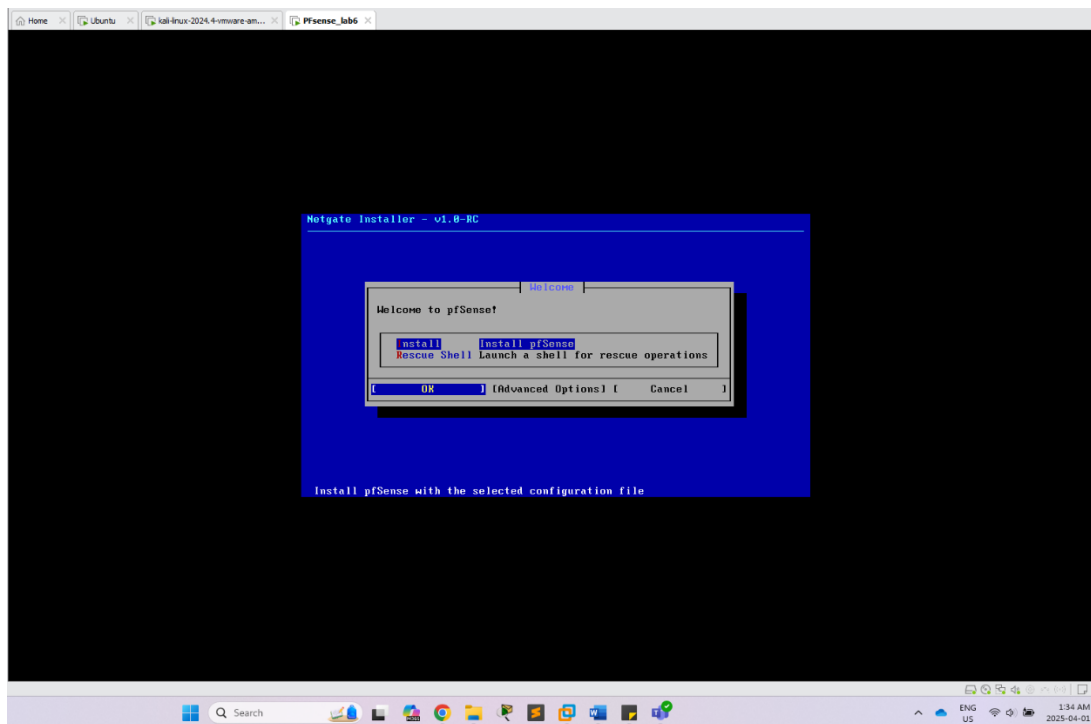
1. Install pfsense on VMware, take screenshots of installation process. **(Attach at least 10 screenshots)**



Lab 6: Firewall & Intrusion Detection Systems (Individual Lab)

INFT 1202

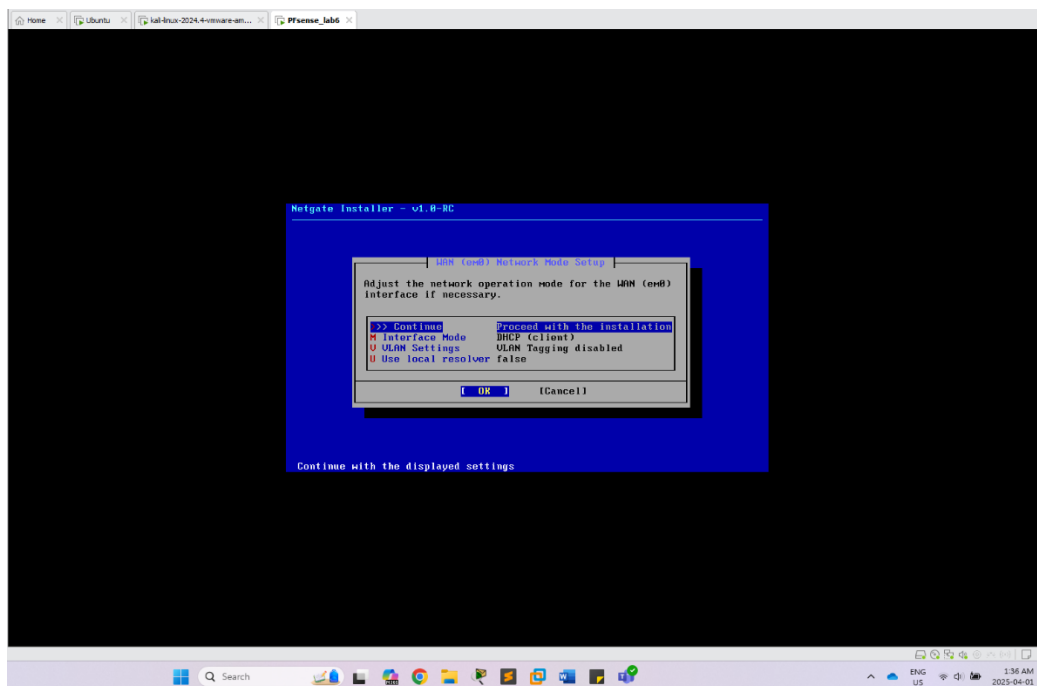
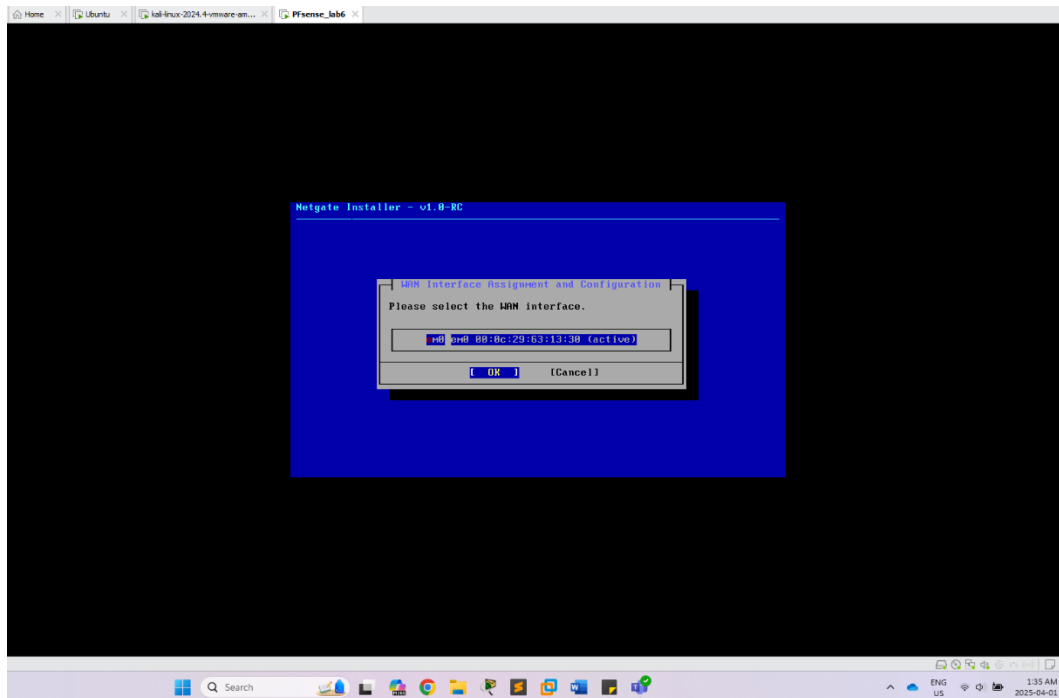
Total Marks :20



Lab 6: Firewall & Intrusion Detection Systems (Individual Lab)

INFT 1202

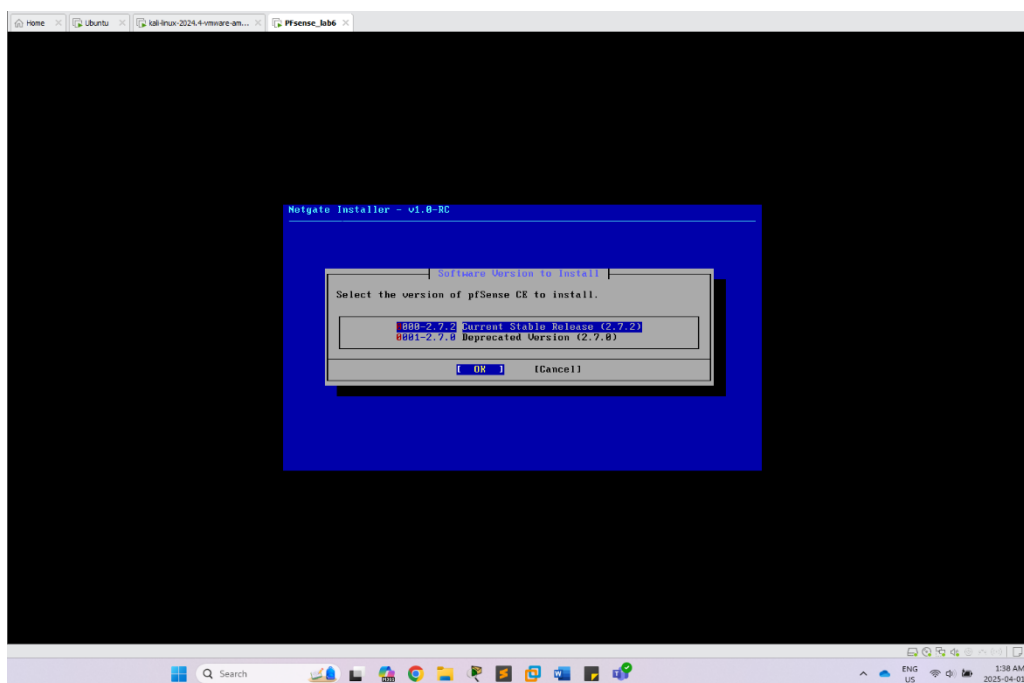
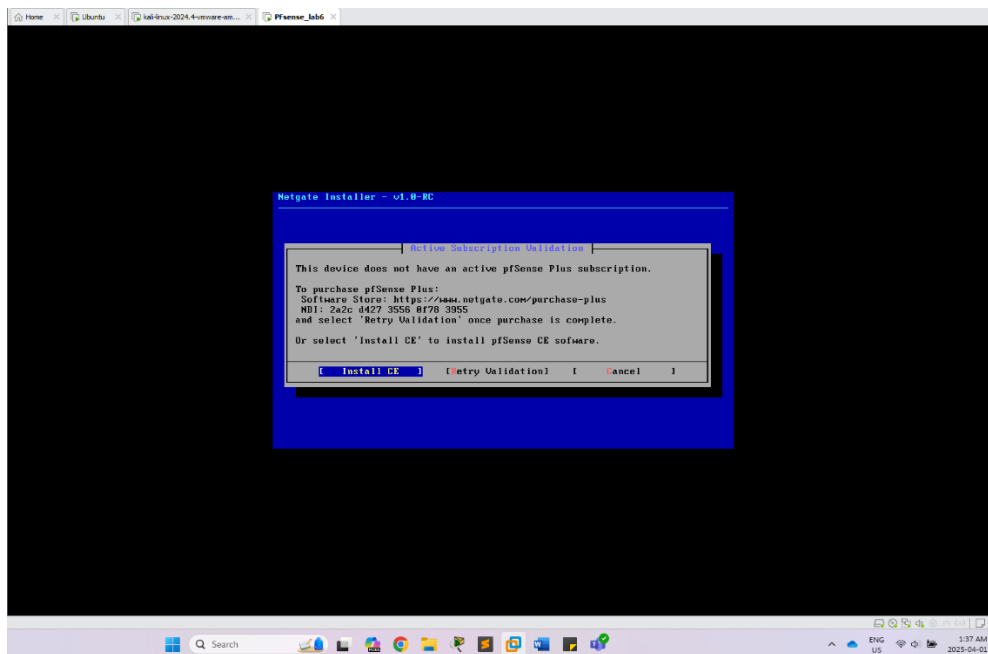
Total Marks :20



Lab 6: Firewall & Intrusion Detection Systems (Individual Lab)

INFT 1202

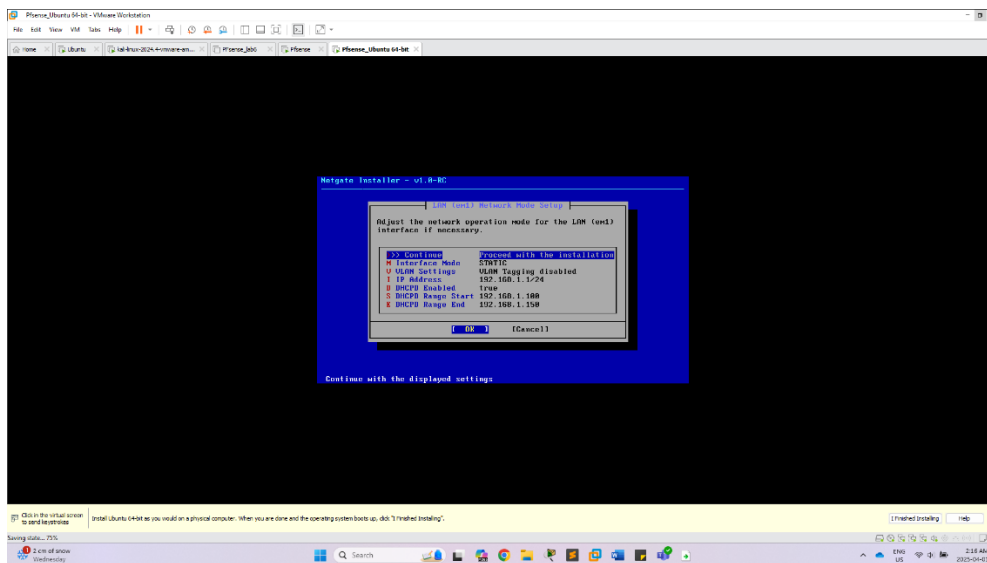
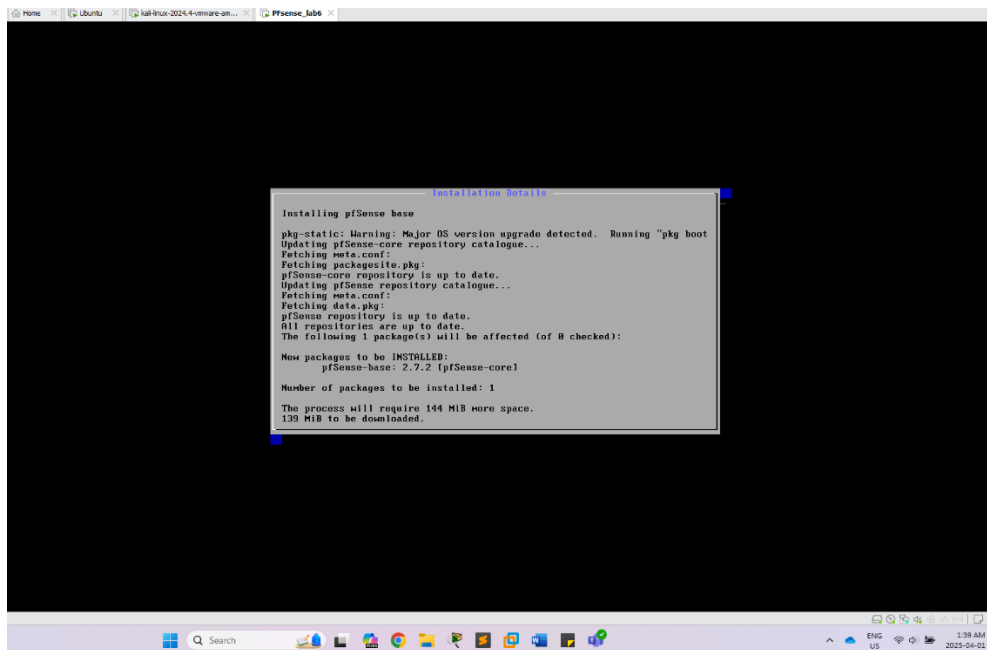
Total Marks :20



Lab 6: Firewall & Intrusion Detection Systems (Individual Lab)

INFT 1202

Total Marks :20



Lab 6: Firewall & Intrusion Detection Systems (Individual Lab)

INFT 1202

Total Marks :20

```
Starting syslog... done.
Starting DSM... done.
pSense 2.7.2 RELEASE amd64 28040304 1353
Bootup complete.

FreeBSD/amd64 (pSense.hsm.org) (ttyu0)

Ubuntu Virtual Machine - Netgate Device ID: 433d7671f908a93b7f56

*** Welcome to pSense 2.7.2 RELEASE (amd64) on pSense ***

WAN (wan) -> em0 -> v4: DHCP4: 192.168.81.152/24
LAN (lan) -> em1 -> v4: 192.168.1.1/24

0) Logon (SSH only)          9) pTop
1) Reassign interface        10) Filter Logs
2) Set interface(s) IP address 11) Restart webConfigurator
3) Reset webConfigurator password 12) PHP Shell & pSense tools
4) Reset to factory defaults 13) Update from remote
5) Reboot system            14) Enable Secure Shell (sshd)
6) Mail system              15) Restore recent configuration
7) Ping host                16) Restart PHP-FPM
8) Shell

Enter an option: █

Ubuntu Virtual Machine - Netgate Device ID: 433d7671f908a93b7f56

*** Welcome to pSense 2.7.2 RELEASE (amd64) on pSense ***

WAN (wan) -> em0 -> v4: DHCP4: 192.168.81.152/24
LAN (lan) -> em1 -> v4: 192.168.1.1/24

0) Logon (SSH only)          9) pTop
1) Reassign interface        10) Filter Logs
2) Set interface(s) IP address 11) Restart webConfigurator
3) Reset webConfigurator password 12) PHP Shell & pSense tools
4) Reset to factory defaults 13) Update from remote
5) Reboot system            14) Enable Secure Shell (sshd)
6) Mail system              15) Restore recent configuration
7) Ping host                16) Restart PHP-FPM
8) Shell

Enter an option: 2

Available interfaces:
1 - WAN (wan - dhcp, dhcp0)
2 - LAN (lan - static)

Enter the number of the interface you wish to configure: █
```


Lab 6: Firewall & Intrusion Detection Systems (Individual Lab)

INFT 1202

Total Marks :20

```
6) Halt system          15) Restore recent configuration
7) Ping host            16) Restart PHP-FPM
8) Shell

Enter an option: 2

Available interfaces:

1 - LAN (em0 - dhcp, dhcp6)
2 - LAN (em1 - static)

Enter the number of the interface you wish to configure: 2

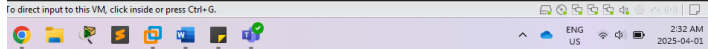
Configure IPv4 address LAN interface via DHCP? (y/n) n

Enter the new LAN IPv4 address. Press <ENTER> for none:
> 192.168.5.1

Subnet masks are entered as bit counts (as in CIDR notation) in pfsense.
e.g. 255.255.255.0 = 24
     255.255.0.0   = 16
     255.0.0.0     = 8

Enter the new LAN IPv4 subnet bit count (1 to 32):
> 
```

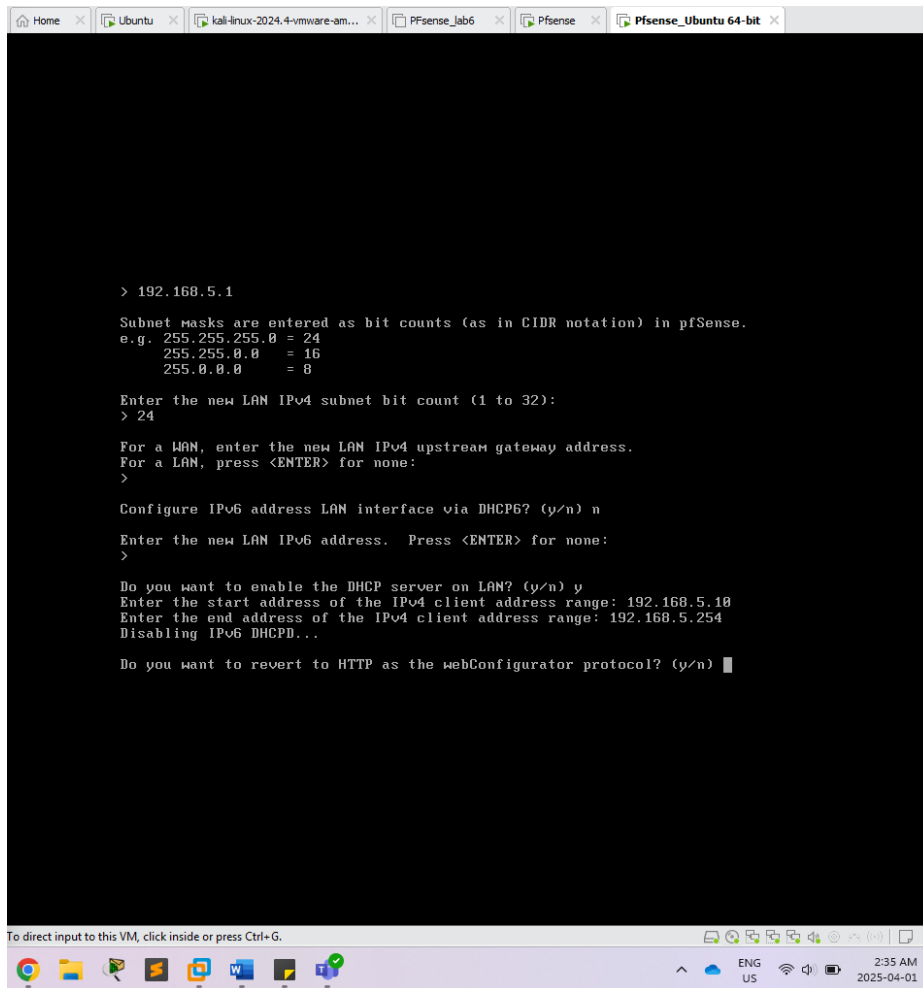
To direct input to this VM, click inside or press Ctrl+G.



Lab 6: Firewall & Intrusion Detection Systems (Individual Lab)

INFT 1202

Total Marks :20



The screenshot shows a terminal window titled 'PfSense_Ubuntu 64-bit' with the following text:

```
> 192.168.5.1

Subnet masks are entered as bit counts (as in CIDR notation) in pfSense.
e.g. 255.255.255.0 = 24
     255.255.0.0   = 16
     255.0.0.0     = 8

Enter the new LAN IPv4 subnet bit count (1 to 32):
> 24

For a WAN, enter the new LAN IPv4 upstream gateway address.
For a LAN, press <ENTER> for none:
>

Configure IPv6 address LAN interface via DHCP6? (y/n) n

Enter the new LAN IPv6 address. Press <ENTER> for none:
>

Do you want to enable the DHCP server on LAN? (y/n) y
Enter the start address of the IPv4 client address range: 192.168.5.10
Enter the end address of the IPv4 client address range: 192.168.5.254
Disabling IPv6 DHCPD...

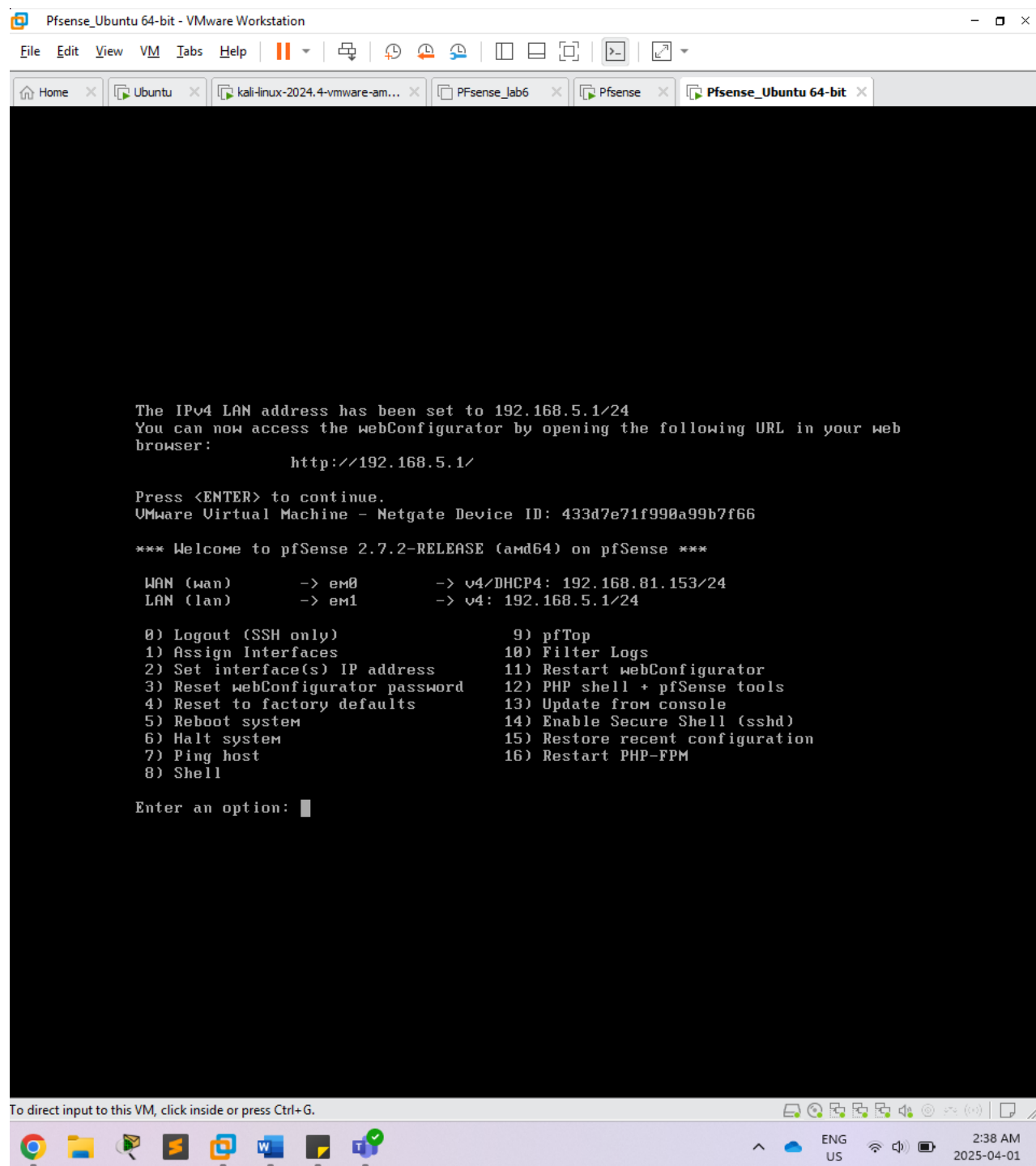
Do you want to revert to HTTP as the webConfigurator protocol? (y/n) █
```

The terminal window is part of a virtual machine environment, as indicated by the taskbar at the bottom showing various application icons and system status information (2:35 AM, 2025-04-01).

Lab 6: Firewall & Intrusion Detection Systems (Individual Lab)

INFT 1202

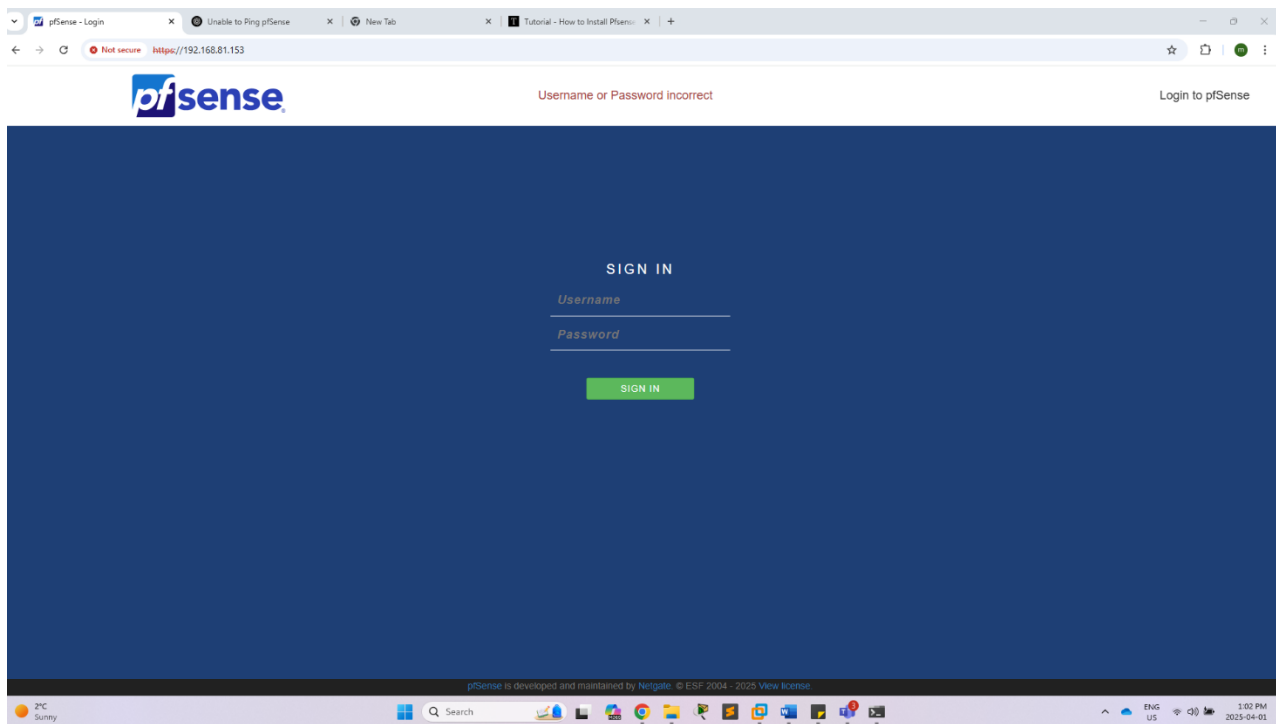
Total Marks :20



Lab 6: Firewall & Intrusion Detection Systems (Individual Lab)

INFT 1202

Total Marks :20

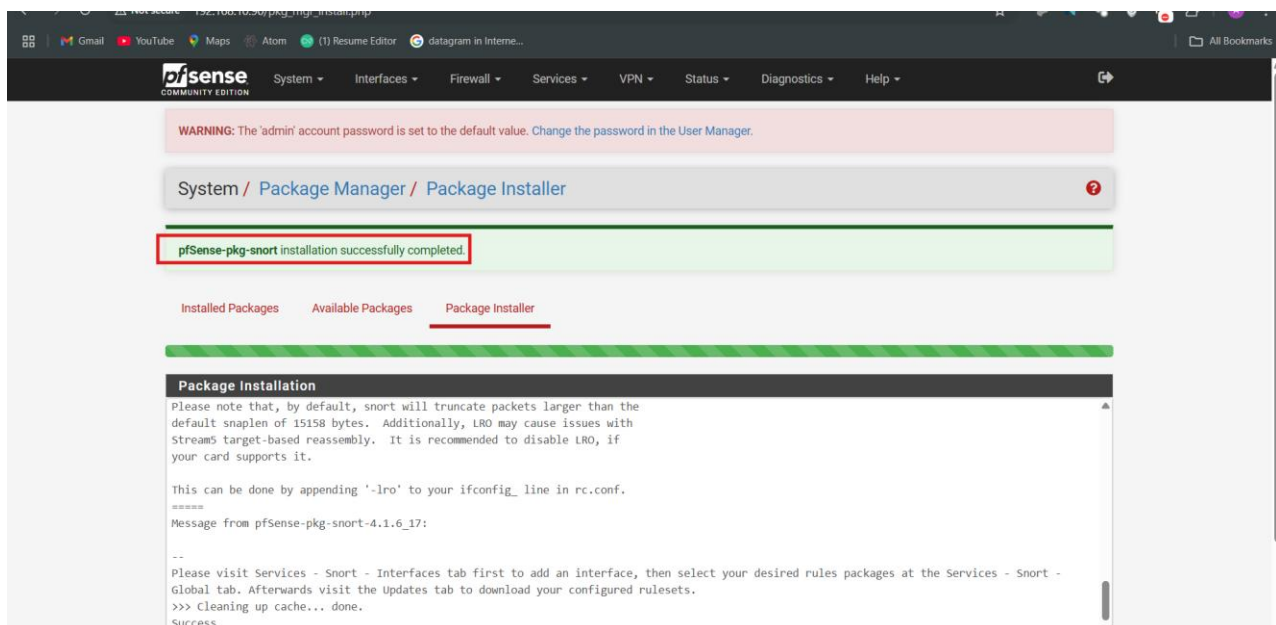
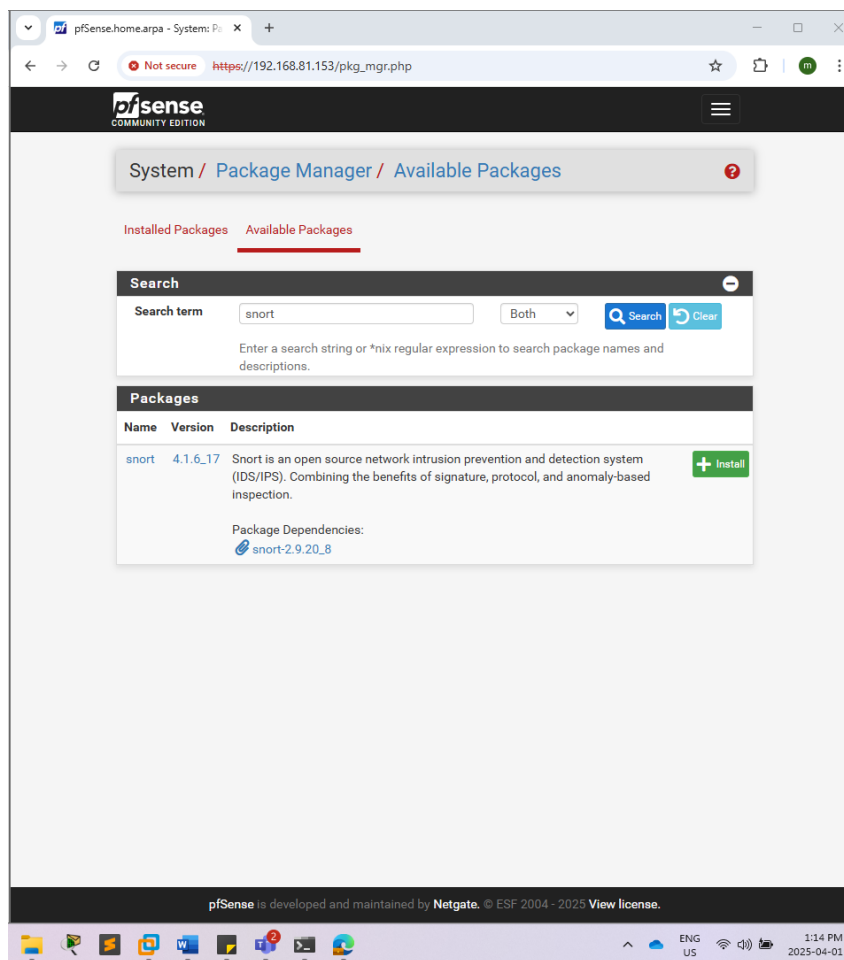


2. Install Snort IDS on your pfsense firewall **(Attach at least 2 screenshots)**

Lab 6: Firewall & Intrusion Detection Systems (Individual Lab)

INFT 1202

Total Marks :20



Part 3

Questions

Answer the questions in the Introduction section and justify your answers.

1. Simple yes or no answer will not get any credits.

- a) What is a zero-day attack?

A zero-day attack occurs when attackers exploit a previously unknown vulnerability in a particular system. That vulnerability was overlooked by everyone, including the system developer/manufacturer.

- b) Can Snort catch zero-day network attacks? If not, why not? If yes, how?

Snort cannot detect zero-day attacks directly. Its IDPS functionalities are based on signatures and rules. However, by establishing strong restrictions, we can avoid the additional damage that a zero-day attack may inflict.